

```
In [2]: pip install pandas
import pandas as pd

Cell In[2], line 1
      pip install pandas
      ^
SyntaxError: invalid syntax

In [3]: pip install pandas

Requirement already satisfied: pandas in /opt/anaconda3/lib/python3.14/site-packages (3.0.3)
Requirement already satisfied: numpy>=2.3.3 in /opt/anaconda3/lib/python3.14/site-packages (from pandas) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/anaconda3/lib/python3.14/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /opt/anaconda3/lib/python3.14/site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

In [4]: import pandas as pd

In [5]: data=pd.read_csv("/Users/tanishqsharma/Documents/trends.csv")

In [6]: print(data.head())

   location  year  category rank  query
0   Global  2001  Consumer Brands    1  Nokia
1   Global  2001  Consumer Brands    2   Sony
2   Global  2001  Consumer Brands    3   BMW
3   Global  2001  Consumer Brands    4   Palm
4   Global  2001  Consumer Brands    5  Adobe

In [7]: ata.shape()

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NameError                                Traceback (most recent call last)
Cell In[7], line 1
----> 1 ata.shape()

NameError: name 'ata' is not defined

In [8]: data.shape()

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TypeError                                Traceback (most recent call last)
Cell In[8], line 1
----> 1 data.shape()

TypeError: 'tuple' object is not callable

In [9]: data.shape

Out[9]: (26955, 5)

In [10]: data.isnull().sum()

Out[10]: location      0
year              0
category          0
rank              0
query            0
dtype: int64

In [11]: data.duplicated().sum()

Out[11]: np.int64(10)

In [12]: data["location"].str.strip()
data["category"].str.strip()
data["query"].str.strip()

Out[12]: 0                Nokia
1                Sony
2                BMW
3                Palm
4                Adobe
...
26950      Virus Corona là gì
26951      Miễn thị thực là gì
26952      Đầu cắt moi là gì
26953      Bệnh bạch hầu là gì
26954      Đông Lào là gì
Name: query, Length: 26955, dtype: str

In [13]: data["year"].astype(int)
data["rank"].astype(int)

Out[13]: 0      1
1      2
2      3
3      4
4      5
..
```

```
26950    1
26951    2
26952    3
26953    4
26954    5
Name: rank, Length: 26955, dtype: int64
```

```
In [14]: data.columns.str.lower()
```

```
Out[14]: Index(['location', 'year', 'category', 'rank', 'query'], dtype='str')
```

```
In [15]: data.shape
```

```
Out[15]: (26955, 5)
```

```
In [16]: data.isnull().sum()
```

```
Out[16]: location    0
year              0
category          0
rank              0
query             0
dtype: int64
```

```
In [17]: data.duplicated().sum()
```

```
Out[17]: np.int64(10)
```

```
In [18]: print(data.head())
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe